COM5231 · Introduction to Wireless and Ad Hoc Networking

Part 8 — Revision Guide

Priorities · Definitions · Formulas · Repeated Questions · Mistakes · Strategy · Checklist

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

Part 8 — Final Examination Revision Guide

Use this as your **last-week and night-before** document. Every item points back to the full notes (Parts 1–3) and answers (Parts 5–6).

A. High-priority topics (from the counted patterns in Part 4)

Priority	Topic	Why	Where
★★★	**Bluetooth Q-bank**: profiles, master vs AP, two piconets, hidden terminal, scatternet probability	Exam Q4 (20 marks) + Tests; profiles asked **4 times**	Part 3 §10; Part 5 Q4
★★★	**802.11 MAC scenarios**: hidden node and CSMA/CA (Figure 2), ACKs, two ISPs channel 11 / 1 + 11	Exam Q5 (18 marks); ISP question asked **3 times**	Part 3 §9; Part 5 Q5
★★★	**Routing**: Dijkstra (Figure 1 from any node; new graphs), LS vs DV, Dijkstra vs BF, count-to-infinity, MANET categories	Exam Q2(d)+Q3 = **30 marks**	Part 3 §11; Part 5 Q2–3
★★★	**Error coding calculations**: Hamming, CRC, checksum; parity vs bursts	In **every** paper except T1A	Part 2 §8
★★	**Wireless channel**: path loss vs multipath vs interference; three propagation mechanisms; propagation problems; interference troubleshooting; distance calculation	Test 1 core; Exam Q1(a)	Part 1 §2
★★	**Spread spectrum and CDMA**: FHSS/DSSS principles; CDMA anti-jamming	Asked **3 times each**	Part 2 §5–6
★★	**OFDM/OFDMA**: three subcarrier approaches; OFDM vs OFDMA	Exam Q1(c), T1A, T1B	Part 2 §7
★★	**Capacity and MIMO**: three ways to increase capacity; SU vs MU-MIMO; mobile node moving away	T1B, Exam Q5(a)	Part 1 §3
★	Nyquist/Shannon, modulation, FDMA/TDMA formulas, generations, 802.15.3/.4, ZigBee, packet formats	In notes, never yet examined	Parts 1–3

B. Essential definitions (reproduce these word-perfect)

Term	Definition
Channel capacity	The maximum rate at which data can be transmitted over a given communication path, or channel, under given conditions
Bandwidth (effective)	The narrow band of frequencies in which most of the signal's energy is contained
SNR	Ratio of signal power to noise power at a point in the transmission; SNR_dB = 10 $\log_{10}(S/N)$
Free space loss	Attenuation because the signal spreads over an ever-larger sphere with distance: $Pt/Pr = (4\pi d/\lambda)^2$
Path loss exponent	Parameter describing how fast received power decreases with distance in an environment (n = 2 in free space)
Reflection	Occurs when a wave meets a surface **large** relative to its wavelength

Term	Definition
Diffraction	Occurs at the **edge** of an impenetrable body large compared to the wavelength; secondary waves reach shadowed areas
Scattering	Occurs when obstacles are **on the order of the wavelength or smaller**; the signal splits into many weaker signals
Multipath	Multiple copies of a signal with varying delays received due to reflection/diffraction/scattering
ISI	Delayed copies of a pulse arrive with the primary pulse of a subsequent bit
Fading	Variation in received signal strength over time/position due to multipath, shadowing and movement
Flat vs frequency-selective fading	Flat: all frequency components fluctuate in the same proportion ($B_c \gg B_s$). Selective: only part of the band is attenuated
Fast vs slow fading	Fast: coherence time not much greater than bit time. Slow: $T_c \gg T_b$
FEC	Receiver corrects errors using only information in the received transmission (redundant bits)
Diversity	Providing multiple independent copies of the signal, because channels fade independently
MIMO	Multiple antennas at transmitter and receiver, used for diversity, beamforming, MU-MIMO, multilayer transmission
AMC	Dynamically adjusting modulation and coding rate to current channel conditions
Multiple access	Many users sharing finite spectrum simultaneously and efficiently, to maximise capacity
FDMA / TDMA / CDMA / SDMA	Divide by frequency (+ guard bands) / time slots / codes over the whole band / space (beams, sectors, reuse)
Spread spectrum	Expanding bandwidth B to $B_{ss} \gg B$ with a pseudorandom code independent of the signal, for privacy and anti-jamming
FHSS	Carrier hops among M frequencies selected by a pseudorandom k-bit pattern each hop period
DSSS	Each data bit is replaced by n chips; chip rate = n × bit rate
OFDM	Multicarrier modulation: data split into N lower-rate streams on orthogonal subcarriers, all from one source
OFDMA	FDMA + TDMA on OFDM: users get subsets of subcarriers (subchannels) at different times
SC-FDMA	OFDMA-like scheme with an extra DFT; lower PAPR; LTE uplink
Single-bit / burst error	One bit changed / two or more bits changed; burst length from first to last bad bit
BSS / ESS	Stations + optional AP / two or more BSSs with APs joined by a distribution system
CSMA/CA	Sense, wait DIFS, random backoff, optional RTS/CTS, data, ACK
NAV	Timer from RTS/CTS durations telling stations how long to wait before checking the channel
Hidden station	Two stations out of each other's range both transmit to a common receiver, causing a collision there
Exposed station	A station refrains from transmitting although its transmission would not cause a collision
Piconet / scatternet	1 master + ≤ 7 active slaves / overlapping piconets sharing area and bandwidth

Term	Definition
Profile	A set of protocols implementing a particular Bluetooth application; a standard of interoperability
Routing / forwarding / switching	Network-wide path selection / router-local input→output move by destination IP / layer-2 frame move by MAC address
Count-to-infinity	DV routing loop where estimates rise slowly after bad news ("bad news travels slowly")

C. Formulas and procedures

C1. Formula sheet

Formula	Use when
$T = 1/f$; $\lambda = c/f$ ($c = 3 \times 10^8$ m/s)	Period, wavelength
$C = 2B \log_2 M$ (Nyquist)	Max bit rate, **noiseless** channel; or find M
$C = B \log_2(1 + SNR)$ (Shannon)	Max bit rate, **noisy** channel. **SNR as a ratio**
$SNR_{dB} = 10 \log_{10}(S/N)$; S/N = 10^(dB/10)	dB conversion
$P_t/P_r = (4\pi d/\lambda)^2 = (4\pi fd/c)^2$	Free-space loss, power, distance, frequency
$P_r = P_t/(4\pi d/\lambda)^2$; $d = (\lambda/4\pi)\sqrt{(P_t/P_r)}$; $f = c\sqrt{L_p/(4\pi d)}$	Rearrangements
$L_{dB} = 20 \log f + 20 \log d - 147.56$	Free-space loss in dB (f in Hz, d in m)
SNR = Pr/Pn; Lp = Pt/Pr	WCP equations 2.2–2.3
$T_c = 0.423c/(vf)$	Coherence time
$N = (B_t - 2B_g)/B_c$	FDMA channels
Users = channels × slots per channel	TDMA users
$S = N/r$, $r = \log_2 L$	Baud vs bit rate
$f_i = f_c + (2i - 1 - M)f_d$	MFSK frequencies
$2^r \geq m + r + 1$	Hamming check bits
P(collision) = 1 − (1 − 1/F)^(2k)	Scatternet: F frequencies, k other piconets, 2 slots per poll+reply (k = 1, F = 80 → 2.5%)
$r_i / \bar{r}_i$	Proportional fairness metric
$d_x(y) = \min_v\{c(x,v) + d_v(y)\}$	Bellman-Ford

C2. Procedures (step lists)

Hamming (even parity, 8 data bits)

1. Inequality: r = 4 (16 ≥ 13).
2. Positions 1–12; P at 1, 2, 4, 8; data in 3, 5, 6, 7, 9, 10, 11, 12 left to right.
3. P1: 3, 5, 7, 9, 11 · P2: 3, 6, 7, 10, 11 · P4: 5, 6, 7, 12 · P8: 9, 10, 11, 12. Each = XOR of its bits (make the group's 1-count even).
4. Write the 12-bit codeword. To correct: recompute checks, read C8C4C2C1 as the bad-bit position.

CRC

1. G bits from the polynomial (degree r = length − 1).
2. Append r zeros.
3. XOR-divide: whenever the leading bit is 1, XOR with G; otherwise bring down.
4. The last r bits = remainder = CRC; transmit data + CRC.
5. Receiver: divide the whole frame; remainder 0 = OK, non-zero = error.

Internet checksum (k-bit words)

1. Add words one at a time; if a carry exits the top bit, **wrap it around** (add 1).
2. Checksum = **complement** of the final sum.
3. Receiver: add everything including the checksum → **all 1s** = OK.

Dijkstra

1. Table: Step | N' | D(n),p(n) for every node.
2. Initialise with the source's neighbours; others ∞.
3. Repeat: add the min-D node to N'; relax its neighbours (D = min(D, D(w) + c)).
4. Show every row; mark settled nodes "—".
5. Finish with a table of cost + full path (+ first hop). **Mention ties.**

Free-space problems

1. Convert units. 2. λ = c/f. 3. Choose the rearrangement. 4. Compute carefully (square root before dividing). 5. Units and a sanity check.

D. Frequently repeated questions: exam-ready short answers

D1. Profiles as an advantage (asked 4×). Interoperability across vendors (the stated purpose of profiles); a ready-made recipe of mandatory/optional features for each usage model; reuse of protocols and other profiles (e.g. File Transfer uses OBEX/RFCOMM/L2CAP + Generic Access); devices implement only the profiles they need (cheap, low power); consistent discovery, connection and security via the Generic Access Profile and SDP.

D2. Path loss vs multipath vs interference (3×). Path loss: deterministic power decrease with distance and frequency, $(4\pi d/\lambda)^2$; fix with power, antennas, more APs. Multipath: own signal arrives via reflection/diffraction/scattering with different delays; random fading and ISI; fix with equalization, diversity, OFDM. Interference: other transmitters (co-/adjacent-channel, microwave ovens); fix with channel planning, spread spectrum, directional antennas.

D3. FHSS and DSSS (3×). SS expands B → Bss ≫ B with a pseudorandom code independent of the signal. FHSS: PN generator → frequency table → synthesizer; the carrier hops among $M = 2^k$ frequencies every hop period; a jammer hits only one hop. DSSS: each bit → n chips (Barker 11: 10110111000); bandwidth × n; privacy and interference immunity.

D4. CDMA and jamming (3×). All users share the whole band all the time, separated by unique pseudorandom codes at the chip rate; the correlator despreads the wanted signal; others stay as noise; power control. Hard to jam: wide band and low power density; despreading spreads and filters out the jammer; the code is unknown; designed for anti-jamming (WWII).

D5. Two ISPs on channel 11 / channels 1 and 11 (3×). Ch 11: no breakdown; one shared collision domain; CSMA/CA makes them defer to each other; simultaneous transmissions collide, no ACK, backoff and retransmit; each gets about half the throughput. Ch 1 + 11: non-overlapping, independent; simultaneous transmissions succeed; each gets the full channel.

D6. CRC (3×). See procedure C2. Past answers: 101110/1001 → **011**; 110101/1011 → **111**; 10011010/1101 → **101** (leftmost flip → remainder 101 ≠ 0).

D7. Master of two piconets (3×). No: the master's address and clock define the hop sequence and timing, so both would be the same piconet. A device can be slave/slave or master/slave.

D8. Scatternet probability (3×). 1/80 per slot; poll + reply need 2 clean slots: $1 - (79/80)^2 \approx$ **2.5%**.

D9. Master vs AP (3×). Master polls (no contention), sets the hop sequence, ≤ 7 slaves, relays inside the PAN. AP: stations contend with CSMA/CA (PCF optional), and the AP bridges to the wired network / other networks.

D10. Three propagation mechanisms (2×). Reflection (large surfaces; ground reflection may cancel LOS); diffraction (sharp edges; reaches shadow areas); scattering (small objects; many weak signals, unpredictable). Without LOS, diffraction and scattering dominate.

D11. OFDMA approaches (2×). Adjacent: equal SINR, choose the best block per user (needs full CSI). Regular: frequency diversity, low estimation burden. Random: diversity + reduced adjacent-cell interference.

D12. Hidden node, Figure 2 (2×). A and C can't hear each other but both reach B; both sense idle, transmit, collide at B; undetected; no ACKs; backoff and retransmit; capacity lost.

D13. CSMA/CA, Figure 2 (2×). A: NAV/sense → DIFS → backoff → RTS → B: CTS (C hears it, sets NAV) → A: DATA → B: ACK (each after SIFS). No CTS/ACK → backoff and retry.

D14. ACKs (2×). Ethernet detects collisions; no collision = implicit success. 802.11 can't detect collisions (half duplex, hidden stations, fading) and has high error rates, so it needs explicit ACKs.

D15. Mobile node moving away (2×). Increase transmit power; lower the rate / more robust modulation and coding (AMC).

D16. Interference troubleshooting (2×). Characterise → spectrum survey → locate the source → remove/relocate/shield → change channel/band → better antenna placement, directional antennas → raise SINR → robust techniques (SS, FEC, AMC) → more APs → verify.

D17. Hamming (2×). 10101111 → **101001001111**; 11001010 → **001110001010**.

D18. Checksum (2×). 1001 1100 1010 0011 → **1011**; 1010 1100 0110 1001 → **1000** (receiver sum 1111).

D19. Parity (2×). Detects all single-bit (odd) errors; misses even-numbered errors (bursts); cannot correct.

D20. MANET categories (2×). Proactive/table-driven (DSDV, OLSR); reactive/on-demand (AODV, DSR); hybrid (ZRP). Hard because of mobility, no infrastructure, bandwidth, energy, unreliable links.

D21. Dijkstra Figure 1 (2×). From x: v3 u6 w6 y6 t7 z8. From t: u2 v4 w5 ×7 y7 z15. From w: u3 v4 t5 ×6 y12 z14.

D22. Count-to-infinity (2×). DV loop after bad news, slowly counting up. A cost decrease or a new link: no, because good news travels fast.

D23. Hidden terminal in Bluetooth (2×). No within a piconet: master polling, slaves transmit only when addressed.

E. Common mistakes (and how to avoid them)

1. **Putting dB into Shannon.** Convert first: 24 dB → 251, not 24.
2. **Mixing Pt in watts with path loss.** Lp = Pt/Pr is a **ratio** (e.g. 125×10^3), not 125×10^{-3} (your notes contain this exact slip).

3. **Forgetting to square-root** when solving $(4\pi d/\lambda)^2 = Lp$ for d, or to **square** when finding Lp.
4. **Units**: mW → 10^{-3}, µW → 10^{-6}, MHz → 10^{6}, km → 10^{3}. Write them every time.
5. **CRC: appending the wrong number of zeros.** Append **degree** = (number of bits in G) − 1, not the number of bits in G.
6. **CRC: doing ordinary subtraction.** It is **XOR**: no borrows.
7. **CRC: writing only the remainder** when asked for the transmitted string. Give **data + remainder**.
8. **Checksum: forgetting the end-around carry**, or forgetting to **complement** the sum.
9. **Hamming: using 3 check bits for 8 data bits.** Always test the inequality: $2^3 = 8 < 12$.
10. **Hamming: mixing up the coverage sets.** P4 covers 4–7 and 12; P8 covers 8–12.
11. **Parity: saying it "corrects" errors or detects all bursts.** It detects only odd numbers of errors and corrects none.
12. **Dijkstra: skipping rows or not showing predecessors.** Marks are for the step-by-step table. Remember G = 5 via E (not 6 direct) in T2-2026, and the **ties** (y from w; C from A).
13. **Dijkstra: forgetting to update** an already-labelled node when a cheaper path appears.
14. **Count-to-infinity: saying a cost decrease causes it.** It doesn't: good news travels fast.
15. **Hidden node: describing the exposed node instead** (or claiming RTS/CTS solves exposed stations).
16. **CSMA/CA: putting DIFS between RTS and CTS.** SIFS separates RTS–CTS–DATA–ACK; DIFS comes before contention.
17. **Two ISPs: saying 802.11 "breaks down".** It doesn't; the channel is shared and collisions increase.
18. **Bluetooth master: claiming it can master two piconets**, or that there are 8 slaves (it's 7 active slaves + 1 master).
19. **Scatternet probability: giving only 1/80.** The question stresses poll **and** reply, so use $1 - (79/80)^2$, and you may mention 1/80 per packet.
20. **OFDM vs OFDMA: treating them as the same.** OFDM = modulation (one user); OFDMA = multiple access (many users).
21. **Saying 4G uses CDMA** (as one table in the notes does). LTE uses **OFDMA/SC-FDMA**.
22. **Three propagation mechanisms: listing "free space, transmission...".** In a **mobile system** the three are **reflection, diffraction, scattering**.

F. Examination answering strategies

1. **Read the mark value and budget**: ~1 developed point per 1–2 marks; ~1 minute per mark (Test 2 2026: 50 marks in 60 minutes).
2. **Definition first**: open long answers with a one-sentence definition. It earns a mark and frames the answer.
3. **Use the lecturer's keywords**: "guard band", "chip rate", "pseudorandom", "NAV", "SIFS/DIFS", "piconet", "FH-TDD-TDMA", "orthogonal", "coherence time". Markers look for them.
4. **Tables for "compare/differences"**: 3–5 rows, then one sentence of conclusion.
5. **Name the nodes in scenarios**: "A sends RTS to B; C hears B's CTS and sets its NAV...". Don't just recite generic theory.
6. **Yes/No questions**: state Yes or No in the **first word**, then justify with the mechanism.
7. **Calculations**: write the **formula**, **substitute with units**, show **every** XOR/addition/table row, **box the final answer**, and add a one-line **check** (e.g. remainder 000; receiver sum 1111). State conventions (Hamming positions; tie-breaking).
8. **Dijkstra**: draw the full table; then a final table of cost + path. Mention equal-cost alternatives.

9. **Long essays (10–12 marks)**: intro → 4–6 headed points, each explained with "because..." → short conclusion.
10. **Answer the easy repeated questions first** (profiles, ISP, CRC, Dijkstra) to bank marks, then return to the rest.
11. **Never leave a calculation blank**: the method marks are available even if the arithmetic slips.

G. Final revision checklist

Tick each item when you can do it **from memory**, without notes.

Transmission and channel

- ☐ T = 1/f, λ = c/f; analog vs digital signals; amplifier vs repeater
- ☐ Nyquist and Shannon; the 3–4 MHz, 24 dB → 8 Mbps → M = 16 example
- ☐ Ground / sky / LOS propagation modes with frequency ranges
- ☐ Five propagation mechanisms; **three mobile mechanisms explained**
- ☐ Free-space loss formula; **solve Test 1A Q14 (9.93 m)** and the WCP problems
- ☐ Path loss exponent table; four noise types
- ☐ Multipath effects; fading types (fast/slow, flat/selective, large/small scale)
- ☐ **Path loss vs multipath vs interference table**
- ☐ Ten wireless problems; **interference troubleshooting steps**; co-channel interference

Correction, capacity and encoding

- ☐ FEC vs ARQ; why ARQ is impractical; equalization; diversity types
- ☐ Four uses of MIMO; **SU-MIMO vs MU-MIMO**
- ☐ AMC; three bandwidth-expansion approaches; **three capacity methods**
- ☐ ASK/FSK/PSK/QPSK/QAM; baud vs bit rate; BER curves

Multiple access, SS, OFDM

- ☐ FDMA/TDMA/CDMA/SDMA: principle, pros, cons; **comparison table**; formulas
- ☐ **CDMA + why hard to jam**
- ☐ **FHSS and DSSS principles**; Barker code; FH-CDMA vs DS-CDMA
- ☐ OFDM orthogonality and benefits; **OFDM vs OFDMA**; **three OFDMA approaches**; proportional fairness; SC-FDMA
- ☐ 1G–5G key facts

Error coding

- ☐ Single-bit vs burst; burst length; **parity capabilities and limits**
- ☐ **Hamming** (inequality, positions, coverage, codeword, syndrome)
- ☐ **CRC** (append, XOR division, receiver check, polynomial ↔ bits)
- ☐ **Internet checksum** (end-around carry, complement, receiver 1111)

WLAN

- ☐ BSS/ESS/AP/SSID; infrastructure vs ad hoc; association
- ☐ Why no CSMA/CD (3 reasons); **CSMA/CA steps**; IFS; NAV; RTS/CTS
- ☐ **Hidden and exposed station** (with Figure 2); **ACKs**; PCF

- ☐ **Two ISPs (channel 11; channels 1 and 11)**; wired vs wireless LAN; design concerns
- ☐ 802.11 PHY variants (a/b/g/n)

Bluetooth / 802.15

- ☐ Stack layers and roles; RFCOMM, TCS BIN, adopted protocols
- ☐ **Profiles advantage**; piconet/scatternet; radio parameters; FH-TDD-TDMA
- ☐ SCO/eSCO/ACL; packet fields; 1/3 FEC, 2/3 FEC, ARQ
- ☐ **Master vs AP; two piconets; hidden terminal in BT; scatternet probability**
- ☐ BT 3.0 HS, BLE; 802.15.3 (PNC), 802.15.4, ZigBee devices and AODV

Routing (supplementary)

- ☐ Routing vs forwarding vs switching
- ☐ **Dijkstra tables** (Figure 1 from x, t, w, u; T2-2026 from A)
- ☐ LS vs DV; Dijkstra vs Bellman-Ford; Bellman-Ford equation
- ☐ **Count-to-infinity**, plus the cost-decrease and new-link cases
- ☐ Why MANET routing is hard; **proactive/reactive/hybrid with examples**

Good luck. If you can tick every item, you can answer every question that has appeared in your past papers.